

Problem Statment

Develop a C++ program to maintain train passenger records using a linked list. The program should allow adding, deleting, searching, and displaying passenger information.

Algorithm: Train Passenger Record System Using Linked List

Step 1: Start the program.

Step 2: Define a Passenger node containing:

Passenger ID

Passenger Name

Passenger Age

Pointer next to the next passenger.

Step 3: Initialize head = NULL to indicate that the linked list is empty.

Step 4: Display the main menu:

1. Add Passenger
2. Delete Passenger
3. Search Passenger
4. Display Passengers
5. Exit

Step 5: Read the user's choice.

Case 1: Add Passenger

1. Create a new passenger node dynamically.
2. Enter Passenger ID, Name, and Age.
3. Set newNode->next = NULL.
4. If head == NULL, make the new node the first node by setting head = newNode.
5. Otherwise, traverse the linked list until the last node is reached.
6. Set the last node's next pointer to the new node.
7. Display "Passenger added successfully."
8. Return to the main menu.

Case 2: Delete Passenger

1. Enter the Passenger ID to be deleted.

2. If head == NULL, display "No passenger records found."
3. If the first node contains the given ID:
 - Store head in a temporary pointer.
 - Move head to the next node.
 - Delete the temporary node.
4. Otherwise, traverse the list until the required ID is found or the end is reached.
5. If the passenger is not found, display "Passenger not found."
6. If found: Store the node to be deleted in a temporary pointer.
 - Adjust the previous node's next pointer.
 - Delete the passenger node.
7. Display "Passenger deleted successfully."
8. Return to the main menu.

Case 3: Search Passenger

1. Enter the Passenger ID to search .
2. Start traversal from head.
3. Compare each node's ID with the entered ID.
4. If the ID matches, display the passenger's ID, name, and age.
5. If the end of the list is reached without a match, display "Passenger not found."
6. Return to the main menu.

Case 4: Display Passengers

1. Check whether head == NULL.
2. If the list is empty, display "No passenger records found."
3. Otherwise, start traversal from head.
4. Display the ID, name, and age of each passenger.
5. Move to the next node using next.
6. Continue until NULL is reached.
7. Return to the main menu.

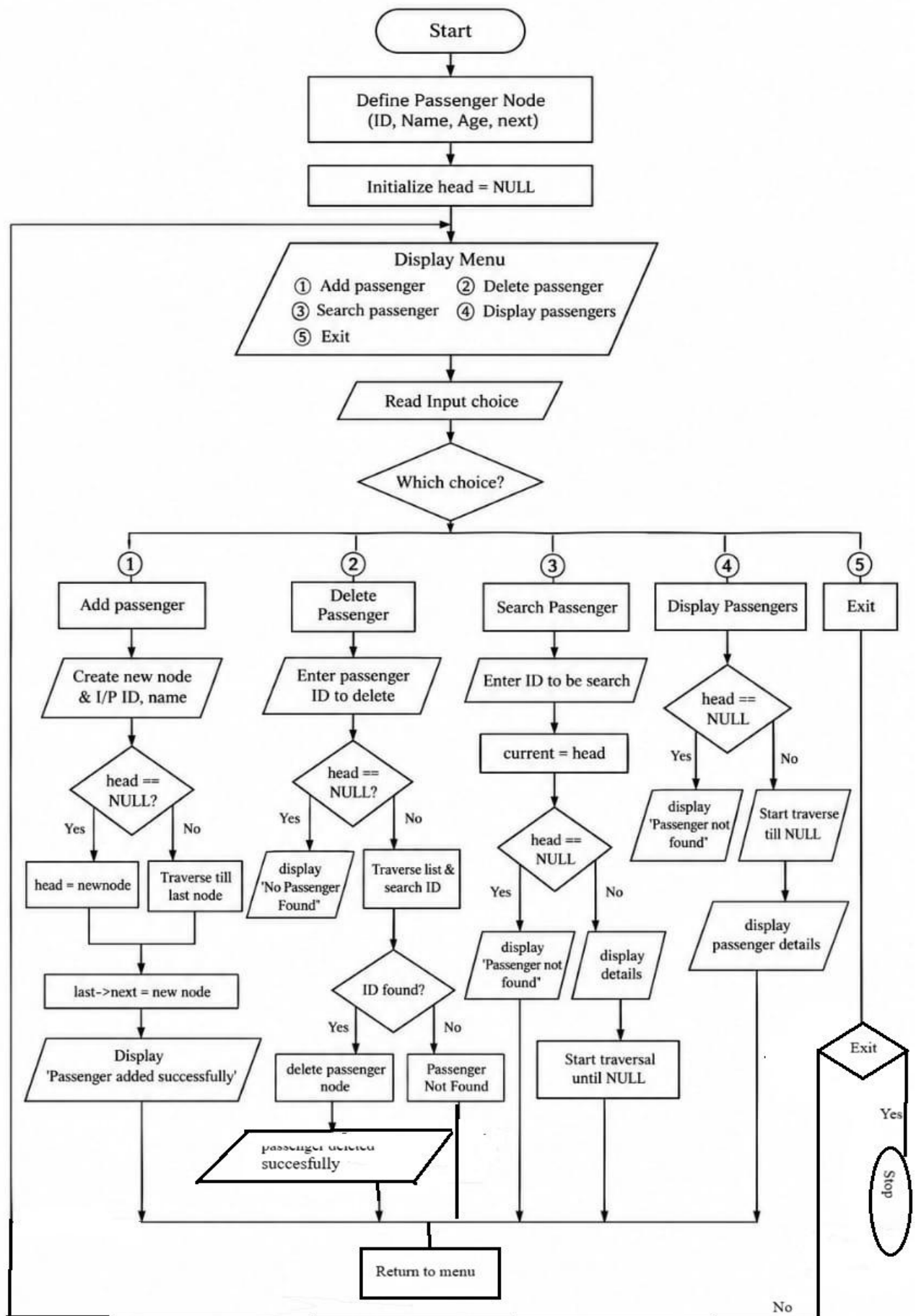
Case 5: Exit

1. Display "Exiting program..."
2. Stop the program.

Step 6: If an invalid choice is entered, display "Invalid choice! Please try again."

Step 7: Repeat Steps 4–6 until the user selects Exit (5).

Step 8: Stop.



C++ Program for Maintaining Train Passenger Record

```
#include <iostream>
#include <string>
using namespace std;
// Node structure
struct Passenger
{
    int id;
    string name;
    int age;
    Passenger* next;
};
Passenger* head = NULL;

// Function to add passenger
void addPassenger()
{
    Passenger* p = new Passenger;
    cout << "\nEnter Passenger ID: ";
    cin >> p->id;
    cout << "Enter Passenger Name: ";
    cin >> p->name;
    cout << "Enter Passenger Age: ";
    cin >> p->age;
    p->next = NULL;
    if (head == NULL)
    {
        head = p;
    }
    else
    {

```

```

    Passenger* temp = head;
    while (temp->next != NULL)
    {
        temp = temp->next;
    }
    temp->next = p;
}
cout << "Passenger added successfully.\n";
}
// Function to delete passenger
void deletePassenger()
{
    int id;
    cout << "\nEnter Passenger ID to delete: ";
    cin >> id;
    if (head == NULL)
    {
        cout << "No passenger records found.\n";
        return;
    }

    // If first passenger is to be deleted
    if (head->id == id)
    {
        Passenger* temp = head;
        head = head->next;
        delete temp;
        cout << "Passenger deleted successfully.\n";
        return;
    }
    Passenger* temp = head;
    while (temp->next != NULL && temp->next->id != id)

```

```

    {
        temp = temp->next;
    }
    if (temp->next == NULL)
    {
        cout << "Passenger not found.\n";
    }
    else
    {
        Passenger* del = temp->next;
        temp->next = del->next;
        delete del;
        cout << "Passenger deleted successfully.\n";
    }
}

// Function to search passenger
void searchPassenger()
{
    int id;
    cout << "\nEnter Passenger ID to search: ";
    cin >> id;
    Passenger* temp = head;
    while (temp != NULL)
    {
        if (temp->id == id)
        {
            cout << "\nPassenger Found!\n";
            cout << "Passenger ID   : " << temp->id << endl;
            cout << "Passenger Name : " << temp->name << endl;
            cout << "Passenger Age  : " << temp->age << endl;
            return;
        }
    }
}

```

```

        temp = temp->next;
    }
    cout << "Passenger not found.\n";
}

// Function to display all passengers
void displayPassengers()
{
    if (head == NULL)
    {
        cout << "\nNo passenger records found.\n";
        return;
    }

    Passenger* temp = head;

    cout << "\n----- Train Passenger Records ----- \n";

    while (temp != NULL)
    {
        cout << "Passenger ID   : " << temp->id << endl;
        cout << "Passenger Name : " << temp->name << endl;
        cout << "Passenger Age  : " << temp->age << endl;
        cout << "----- \n";

        temp = temp->next;
    }
}

// Main function
int main()
{
    int choice;
    do
    {

```

```
cout << "\n===== TRAIN PASSENGER RECORD SYSTEM =====";
cout << "\n1. Add Passenger";
cout << "\n2. Delete Passenger";
cout << "\n3. Search Passenger";
cout << "\n4. Display Passengers";
cout << "\n5. Exit";
cout << "\nEnter your choice: ";
cin >> choice;
switch (choice)

{

    case 1:
        addPassenger();
        break;

    case 2:
        deletePassenger();
        break;

    case 3:
        searchPassenger();
        break;

    case 4:
        displayPassengers();
        break;

    case 5:
        cout << "\nExiting program...\n";
        break;
    default:
```

```
        cout << "\nInvalid choice! Please try again.\n";
    }
} while (choice != 5);
return 0;
}
```

Output :

```
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===== TRAIN PASSENGER RECORD SYSTEM =====
1. Add Passenger
2. Delete Passenger
3. Search Passenger
4. Display Passengers
5. Exit
Enter your choice: 1

Enter Passenger ID: 22771
Enter Passenger Name: rahul
Enter Passenger Age: 20
Passenger added successfully.

===== TRAIN PASSENGER RECORD SYSTEM =====
1. Add Passenger
2. Delete Passenger
3. Search Passenger
4. Display Passengers
5. Exit
Enter your choice: 1

Enter Passenger ID: 22772
Enter Passenger Name: tanmay
Enter Passenger Age: 19
Passenger added successfully.

===== TRAIN PASSENGER RECORD SYSTEM =====
1. Add Passenger
2. Delete Passenger
3. Search Passenger
4. Display Passengers
5. Exit
Enter your choice: 2

Enter Passenger ID to delete: 22771
Passenger deleted successfully.

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===== TRAIN PASSENGER RECORD SYSTEM =====
1. Add Passenger
2. Delete Passenger
3. Search Passenger
4. Display Passengers
5. Exit
Enter your choice: 3

Enter Passenger ID to search: 22772

Passenger Found!
Passenger ID : 22772
Passenger Name : tanmay
Passenger Age : 19

===== TRAIN PASSENGER RECORD SYSTEM =====
1. Add Passenger
2. Delete Passenger
3. Search Passenger
4. Display Passengers
5. Exit
Enter your choice: 4

----- Train Passenger Records -----
Passenger ID : 22772
Passenger Name : tanmay
Passenger Age : 19
-----

===== TRAIN PASSENGER RECORD SYSTEM =====
1. Add Passenger
2. Delete Passenger
3. Search Passenger
4. Display Passengers
5. Exit
Enter your choice: 5

Exiting program...

-----
Process exited after 90.82 seconds with return value 0
Press any key to continue . . . |
```